SC2 Impedance Control - Developer Notes

Lucas M. O. Silva

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Notation

$\boldsymbol{f}$ Dynamic model of the robot.

$\boldsymbol{x}^d$ Desired position vector in the Cartesian space.

$\dot{\boldsymbol{x}}^d$ Desired velocity vector in the Cartesian space.

$\ddot{\boldsymbol{p}}_d$ Desired acceleration vector in the Cartesian space.

$\boldsymbol{q}^d$ Desired position vector in the joint space.

$\dot{\boldsymbol{q}}^d$ Desired velocity vector in the joint space.

$\ddot{\boldsymbol{q}}_d$ Desired acceleration vector in the joint space.

$\boldsymbol{M}_x$ Inertia matrix in the Cartesian space.

$\boldsymbol{N}_x$ Coupling forces (e.g. Coriolis and centrifugal) in the Cartesian space.

$\boldsymbol{G}_x$ Gravity forces in the Cartesian space.

$\boldsymbol{M}$ Inertia matrix in the joint space.

$\boldsymbol{N}$ Coupling torques (e.g. Coriolis and centrifugal) in the joint space.

$\boldsymbol{G}$ Gravity torques in the joint space.

$\boldsymbol{M}^d$ Desired / virtual inertia matrix.

$\boldsymbol{D}^d$ Desired / virtual damping matrix.

$\boldsymbol{K}^d$ Desired / virtual stiffness matrix.

K_F Force feedback gain.

$\boldsymbol{F}_e$ External force vector.

$\boldsymbol{F}_c$ Commanded force vector.

$\boldsymbol{\tau}_e$ External torque vector.

$\boldsymbol{\tau}_c$ Commanded torque vector.

1 Problem definition

The goal of this project is to realize position-based impedance control of a single grapple finger. In other words, given a desired position trajectory in the joint space and joint torque measurements from the actuator, compute the necessary torque command for both joints of the finger according to a target impedance behavior. Furthermore, analyzing the construction of the robot yields the following insights:

- Task space: $\mathbb{R}^{m=2}$, since we only care about the position of the fingertip in the Cartesian plane.
- Workspace: $\mathbb{R}^{n=2}$, due to shoulder and elbow flexion/extension.
- Because $n = m = 2$, the robot is **not** redundant.

2 Cartesian space dynamic model of the robot

The dynamics of the robot can be modeled as

$$\boldsymbol{f}(\ddot{\boldsymbol{x}}, \dot{\boldsymbol{x}}, \boldsymbol{x}, \ddot{\boldsymbol{q}}, \dot{\boldsymbol{q}}, \boldsymbol{q}) = \boldsymbol{F}_c + \boldsymbol{F}_e \,, \tag{1}$$

where $\boldsymbol{f}$ is a non-linear function usually of the form

$$\boldsymbol{f}(\ddot{\boldsymbol{x}}, \dot{\boldsymbol{x}}, \boldsymbol{x}, \ddot{\boldsymbol{q}}, \dot{\boldsymbol{q}}, \boldsymbol{q}) = \boldsymbol{M}_x(\boldsymbol{q})\ddot{\boldsymbol{x}} + \boldsymbol{N}_x(\dot{\boldsymbol{q}}, \boldsymbol{q}) + \boldsymbol{G}_x(\boldsymbol{q}) + \underbrace{\ldots}_{\text{friction, etc.}} \,. \tag{2}$$

We'll come back to this model when we introduce the gravity compensation term to the control law in Section 4. For understanding the basic principle of impedance control, however, Equation (1) suffices.

3 Control law

Let the control law be defined as

$$\boldsymbol{F}_c = (1 + K_F)[\boldsymbol{M}^d(\ddot{\boldsymbol{x}}^d - \ddot{\boldsymbol{x}}) + \boldsymbol{D}^d(\dot{\boldsymbol{x}}^d - \dot{\boldsymbol{x}}) + \boldsymbol{K}^d(\boldsymbol{x}^d - \boldsymbol{x})] + K_F\boldsymbol{F}_e. \tag{3}$$

Plugging this control law into the system dynamics we get

$$\boldsymbol{f}(\ddot{\boldsymbol{x}}, \dot{\boldsymbol{x}}, \boldsymbol{x}, \ddot{\boldsymbol{q}}, \dot{\boldsymbol{q}}, \boldsymbol{q}) = (1+K_F)[\boldsymbol{M}^d(\ddot{\boldsymbol{x}}^d - \ddot{\boldsymbol{x}}) + \boldsymbol{D}^d(\dot{\boldsymbol{x}}^d - \dot{\boldsymbol{x}}) + \boldsymbol{K}^d(\boldsymbol{x}^d - \boldsymbol{x}) + \boldsymbol{F}_e] \tag{4}$$

$$\Leftrightarrow \frac{\boldsymbol{f}(\ddot{\boldsymbol{x}}, \dot{\boldsymbol{x}}, \boldsymbol{x}, \ddot{\boldsymbol{q}}, \dot{\boldsymbol{q}}, \boldsymbol{q})}{1 + K_F} + \boldsymbol{M}^d(\ddot{\boldsymbol{x}} - \ddot{\boldsymbol{x}}^d) + \boldsymbol{D}^d(\dot{\boldsymbol{x}} - \dot{\boldsymbol{x}}^d) + \boldsymbol{K}^d(\boldsymbol{x} - \boldsymbol{x}^d) = \boldsymbol{F}_e. \tag{5}$$

By increasing the force feedback gain K_F, we can drive the dynamics towards the desired impedance behavior $\boldsymbol{M}^d(\ddot{\boldsymbol{x}} - \ddot{\boldsymbol{x}}^d) + \boldsymbol{D}^d(\dot{\boldsymbol{x}} - \dot{\boldsymbol{x}}^d) + \boldsymbol{K}^d(\boldsymbol{x} - \boldsymbol{x}^d) = \boldsymbol{F}_e$ [3] [1]. However, one can't increase K_F indefinitely as this leads to instability. Colgate shows that stability is only guaranteed for $K_F < 1$ [2].

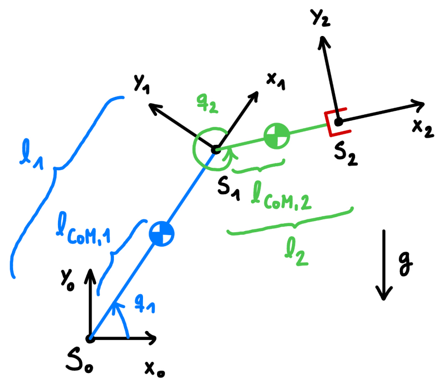

Figure 1: Denavit-Hartenberg convention for a 2 DoF robot arm with rotational joints.

Since the actuators support torque commands, we can use the Jacobian to transform our force command into a torque command

$$\boldsymbol{\tau}_c = (1 + K_F)\boldsymbol{J}^T(\boldsymbol{q})[\boldsymbol{M}^d(\ddot{\boldsymbol{x}}^d - \ddot{\boldsymbol{x}}) + \boldsymbol{D}^d(\dot{\boldsymbol{x}}^d - \dot{\boldsymbol{x}}) + \boldsymbol{K}^d(\boldsymbol{x}^d - \boldsymbol{x})] + K_F\boldsymbol{\tau}_e. \quad (6)$$

Using the forward kinematics and the Jacobian for the desired and measured joint angles, we complete our transition to the joint space. Note that we can even mix Cartesian space trajectories with joint space angle measurements by simply not transforming the desired vectors

$$\boldsymbol{x} = \boldsymbol{FK}(\boldsymbol{q}), \quad (7)$$

$$\dot{\boldsymbol{x}} = \boldsymbol{J}(\boldsymbol{q})\dot{\boldsymbol{q}}, \quad (8)$$

$$\ddot{\boldsymbol{x}} = \dot{\boldsymbol{J}}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{J}(\boldsymbol{q})\ddot{\boldsymbol{q}}. \quad (9)$$

This is the final control law. Every variable is given: $\boldsymbol{q}$ and its derivatives, as well as $\boldsymbol{\tau}_e$, are provided by the sensors in the actuators. $\boldsymbol{q}^d$ and its derivatives are provided by the planner, $\boldsymbol{FK}$, $\boldsymbol{J}$ and $\dot{\boldsymbol{J}}$ are derived from the robot's geometry (see Section 3.1 for more details). Finally, K_F, $\boldsymbol{M}^d$, $\boldsymbol{D}^d$ and $\boldsymbol{K}^d$ are tuning variables.

Because most actuators usually don't provide direct measurements of the joint accelerations, some implementations leave out the inertial term of the impedance equation. This reduces the control law to $\boldsymbol{F}_c = (1 + K_F)[\boldsymbol{D}^d(\dot{\boldsymbol{x}}^d - \dot{\boldsymbol{x}}) + \boldsymbol{K}^d(\boldsymbol{x}^d - \boldsymbol{x})] + K_F\boldsymbol{F}_e$.

3.1 Forward kinematics

Figure 1 shows a depiction of the robot finger, where l_i, $i \in \{1, 2\}$ are the link lengths and q_i, $i \in \{1, 2\}$ are the joint angles according to the Denavit-Hartenberg convention for defining local joint coordinate systems. The forward kinematics are then given as

$$\boldsymbol{x} = \begin{bmatrix} r_x \\ r_y \end{bmatrix} = \boldsymbol{FK}(\boldsymbol{q}) = \begin{bmatrix} l_1 \cos(q_1) + l_2 \cos(q_1 + q_2) \\ l_1 \sin(q_1) + l_2 \sin(q_1 + q_2) \end{bmatrix}. \tag{10}$$

The Jacobian matrix is defined as

$$\boldsymbol{J}(\boldsymbol{q}) = \begin{bmatrix} \frac{\partial r_x}{\partial q_1} & \frac{\partial r_x}{\partial q_2} \\ \frac{\partial r_y}{\partial q_1} & \frac{\partial r_y}{\partial q_2} \end{bmatrix}. \tag{11}$$

Using the forward kinematics shown above, computing the Jacobian generates

$$\boldsymbol{J}(\boldsymbol{q}) = \begin{bmatrix} -l_1 \sin(q_1) - l_2 \sin(q_1 + q_2) & -l_2 \sin(q_1 + q_2) \\ l_1 \cos(q_1) + l_2 \cos(q_1 + q_2) & l_2 \cos(q_1 + q_2) \end{bmatrix}. \tag{12}$$

Taking the time derivative of the Jacobian we get

$$\dot{\boldsymbol{J}}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \begin{bmatrix} -l_1 \cos(q_1)\dot{q_1} - l_2 \cos(q_1 + q_2)(\dot{q_1} + \dot{q_2}) & -l_2 \cos(q_1 + q_2)(\dot{q_1} + \dot{q_2}) \\ -l_1 \sin(q_1)\dot{q_1} - l_2 \sin(q_1 + q_2)(\dot{q_1} + \dot{q_2}) & -l_2 \sin(q_1 + q_2)(\dot{q_1} + \dot{q_2}) \end{bmatrix}. \tag{13}$$

4 Gravity compensation

Gravity compensation is a method for canceling out the effect of gravity on the robot dynamics. Concretely, given a model of the gravitational force (Cartesian space model) or torque (joint space model), adding it as a component of the actuator command nullifies the actual gravitational force or torque to the extent of the model's precision.

Since our control law is defined in the joint space, we'll proceed to analyze the torques caused by gravity. For conservative forces such as the gravitational force, there exists a corresponding potential energy such that

$$\boldsymbol{\tau}_g = \frac{\partial V}{\partial \boldsymbol{q}}. \tag{14}$$

The potential gravitational energy of the robot can be computed from the link masses and lengths to the CoMs m_i, $l_{CoM,i}$, $i \in \{1, 2\}$, as well as the joint angles $\boldsymbol{q}$, as

$$V = m_1 g r_{y,CoM,1} + m_2 g r_{y,CoM,2} \tag{15}$$

$$= m_1 g l_{CoM,1} \sin(q_1) + m_2 g [l_1 \sin(q_1) + l_{CoM,2} \sin(q_1 + q_2)]. \tag{16}$$

With this result, Equation (14) resolves to

$$\boldsymbol{\tau}_g = \begin{bmatrix} m_1 g l_{CoM,1} \cos(q_1) + m_2 g [l_1 \cos(q_1) + l_{CoM,2} \cos(q_1 + q_2)] \\ m_2 g l_{CoM,2} \cos(q_1 + q_2) \end{bmatrix}. \tag{17}$$

From Equation (2) we know that our dynamic model of the robot looks something like this, in joint space

$$\boldsymbol{f}(\ddot{\boldsymbol{q}}, \dot{\boldsymbol{q}}, \boldsymbol{q}) = \boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{N}(\dot{\boldsymbol{q}}, \boldsymbol{q}) + \boldsymbol{G}(\boldsymbol{q}) + \underbrace{\dots}_{\text{friction, etc.}}, \tag{18}$$

and the updated control law now has an additional gravitational torque component as such

$$\boldsymbol{\tau}_c = \underbrace{(1 + K_F)\boldsymbol{J}^T(\boldsymbol{q})\{\dots\} + K_F\boldsymbol{\tau}_e}_{\boldsymbol{\tau}_{impedance}} + \boldsymbol{\tau}_g. \tag{19}$$

If our model is accurate i.e., $\boldsymbol{G}(\boldsymbol{q}) \approx \boldsymbol{\tau}_g$, putting the updated control law in Equation (18) leads to

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{N}(\dot{\boldsymbol{q}}, \boldsymbol{q}) + \boldsymbol{G}(\boldsymbol{q}) + \underbrace{\dots}_{\text{friction, etc.}} = \boldsymbol{\tau}_{impedance} + \boldsymbol{\tau}_g \tag{20}$$

$$\Leftrightarrow \boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{N}(\dot{\boldsymbol{q}}, \boldsymbol{q}) + \underbrace{\dots}_{\text{friction, etc.}} = \boldsymbol{\tau}_{impedance}. \tag{21}$$

Note that the gravity compensation term above **does not** make any considerations about buoyancy i.e., how being underwater introduces additional vertical forces that can effectively lower the forces experienced by the robot. Until a better model is derived, we combat these uncertainties with torque feedback and slow movements.

References

[1] F. Caccavale, C. Natale, B. Siciliano, and L. Villani. Integration for the next generation: embedding force control into industrial robots. *IEEE Robotics & Automation Magazine*, 12(3):53–64, September 2005.

[2] James Edward Colgate. *The control of dynamically interacting systems*. PhD thesis, MIT, 1988.

[3] Neville Hogan and Stephen Buerger. Impedance and Interaction Control. In Thomas Kurfess, editor, *Robotics and Automation Handbook*. CRC Press, October 2004.